Chapter

Light

---

1. An endoscope uses optic fiber to transmit high resolution images of internal organs without loss of information. The principle of light that is used by the optic fiber is based on:

(a) refraction

(b) reflection

(c) scattering

(d) total internal reflection

SQP2025

2. White light is dispersed by a prism. Inside the prism, compared to the blue light, the red light

(a) slows down less and refracts more

(b) slows down more and refracts less

(c) slows down more and refracts more

(d) slows down less and refracts less

SQP2025

3. A convex lens has focal length 12 cm with an object at a distance of 20 cm in front of the lens. He obtains a blurred image on the screen placed at a distance of 23 cm in front of the lens. In order to obtain the clear image, he has to move the screen

(a) towards the lens.

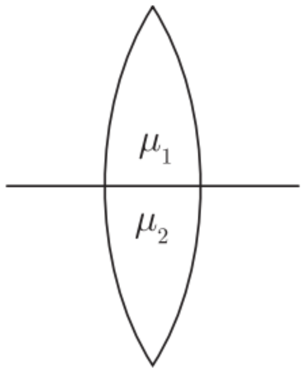
(b) away from the lens.

(c) to a position very far away from the lens.

(d) either towards or away from the lens.

SQP2025

4. Refer to the diagram given below. A lens with two different refractive indices is shown. If the rays are coming from a distant object, then how many images will be seen?



SQP 2025

5. A glass lens always forms a virtual, erect and diminished image of an object kept in front of it. Identify the lens.

SQP 2025

6. When sunlight passes through water droplets in the atmosphere it gets dispersed into its constituent colours forming a rainbow. A similar phenomenon is observed when white light passes through a prism.

(a) Which colour will show the maximum angle of deviation and which colour will show the minimum angle of deviation?

(b) If Instead of sunlight, a green-coloured ray is passed through a glass prism. What will be the colour of the emergent ray?

SQP 2025

7. The diagram below shows a fish in the tank and its image seen in the surface of water.

(a) Name the phenomenon responsible for the formation of this image.

(b) A double convex lens with refractive index μ_1 inside two liquids of refractive indices μ_2 and μ_3 are shown in the diagrams below. The refractive indices are such that $\mu_2 > \mu 1$ and $\mu_1 > \mu_3$.

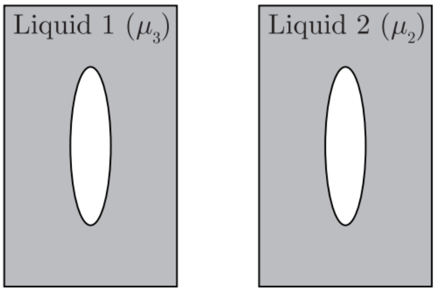


Figure (a) Figure (b)

How would a parallel incident beam of light refract when it comes out of the lens in each of the cases shown above?

(1) in Figure a. (2) in Figure b.

SQP 2025

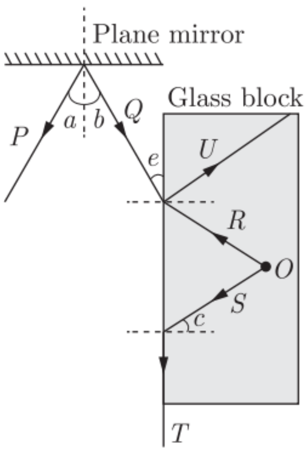
8. The refractive index of water is 1.33 at a certain temperature. When the temperature of water is increased by 40 °C, the refractive index changes to 'x'.

(a) State whether $x < 1.33$ or $x > 1.33$.

(b) State two differences between normal reflection and total internal reflection.

SQP 2025

9. O is a luminescent particle trapped inside a glass block. A student traces the path of rays coming out of it and reflecting over a plane mirror as shown in the diagram below.

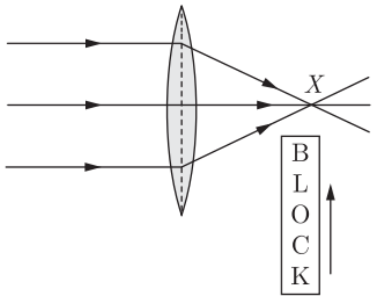


Complete the table, using the labels from the figure.

	Description	Label
a.	an angle of reflection on the mirror	a
b.	a partially reflected ray in the glass slab	
c.	a critical angle	
d.	a refracted ray	
e.	an angle of refraction of the ray R	

SQP 2025

10. A block of glass is pushed into the path of the light as shown below. Then the converging point X will



(a) Move away from the slab

(b) Move towards the slab

(c) Not shift

(d) Move towards the left side of the lens

MAIN 2024

11. Linear magnification(m) produced by a concave lens is

(a) $m < 1$ (b) $m > 1$

(c) $m = 1$ (d) $m = 2$

MAIN 2024

12. Assertion : Ultraviolet radiations are scattered more as compared to the microwave radiations.

Reason : Wavelength of ultraviolet radiation is more than the wavelength of microwave radiation.

(a) Both Assertion and Reason are true.

(b) Assertion is true but Reason is false.

(c) Assertion is false but Reason is true.

(d) Both Assertion and Reason are false.

MAIN 2024

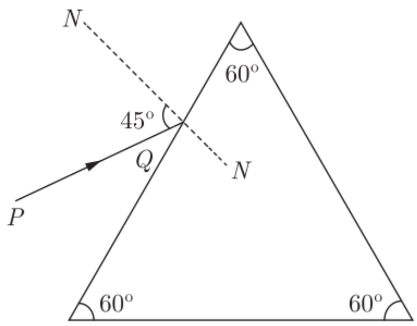
13. In a reading glass what is the position of the object with respect to the convex lens used?

MAIN 2024

14. Why can we not use concave lens for the same purpose?

MAIN 2024

15. A monochromatic ray of light is incident on an equilateral prism placed at minimum deviation position with an angle of incidence 45° as shown in the diagram.



(a) Copy the diagram and complete the path of the ray PQ.

(b) State two factors on which the angle of deviation depends.

MAIN 2024

16. (a) Refractive index of glass with respect to water is $\frac{9}{5}$. Find the refractive index of water with respect to glass.

(b) Name the principle used to find the value in part (a).

(c) If we change the temperature of water, then will the ratio $\frac{9}{8}$ remain the same? Write Yes or No.

MAIN 2024

17. Light travels a distance of '$10x$' units in time 't_1' in vacuum and it travels a distance of 'x' units in time 't_2' in a denser medium. Using this information answer the question that follows:

(a) 'Light covers a distance of '$20x$' units in time 't_1' in diamond. 'State true or false.

(b) Calculate the refractive index of the medium in terms of 't_1' and 't_2'.

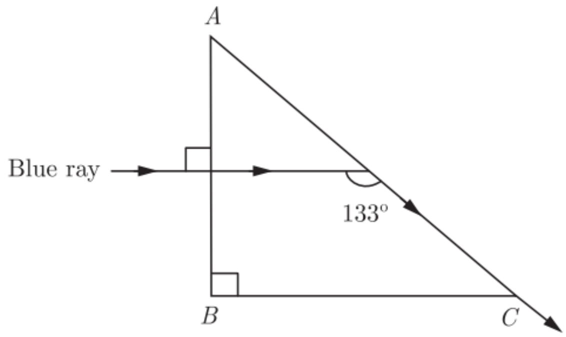
MAIN 2024

18. The diagram below shows the path of a blue ray through the prism:

1. Calculate the critical angle of the material of the prism for blue colour.

2. What is the measure of the angle of this prism (A)?

3. Which colour should replace the blue ray, for the ray to undergo Total Internal Reflection?



MAIN 2024

19. The image of a candle flame placed at a distance of 36 cm from a spherical lens, is formed on a screen placed at a distance of 72 cm from the lens. Calculate the focal length of the lens and its power.

MAIN 2024

20. Optical fibre works on the principle of

(a) total internal reflection

(b) optical transmission

(c) satellite signals

(d) internet mobile towers

SQP 2023

21. A ray of light is incident normally on a plane glass slab. What will be the angle of refraction and the angle of deviation for the ray ?

SQP 2023

22. What rays exist beyond the visible-red end of the electromagnetic spectrum? State one use and one method of detecting these rays.

SQP 2023

23. Draw a ray diagram to illustrate how a ray of light incident obliquely on one face of a rectangular glass slab of uniform thickness emerges parallel to its original direction. Mention which pairs of angles are equal.

SQP 2023

24. Calculate the power of the combination formed by placing two thin lenses of power +5 D and -7 D in contact with each other. State whether the combination will be convergent or divergent.

SQP 2023

25. When a ray of light passes through an equilateral glass prism :

(a) it bends towards the base on both refracting surfaces.

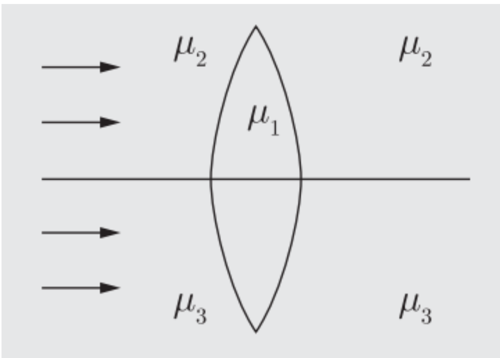
(b) it suffers refraction on the first refracting surfaces.

(c) it suffers refraction on both the refracting surfaces.

(d) both (b) and (c)

MAIN 2023

26. A double convex lens, made of a material of refractive index μ_1, is placed inside two liquids of refractive indices μ_2 and μ_3 as shown $\mu_2 > \mu_1 > \mu_3$. A wide, parallel beam of light is incident on the lens from the left. The lens will give rise to



(a) A single convergent beam

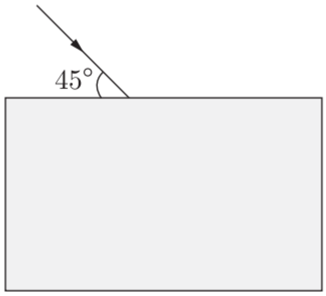
(b) A convergent and a divergent beam

(c) Two different convergent beams

(d) Two different divergent beams

MAIN 2023

27. Draw the diagram given below and clearly show the path taken by the emergent ray.



MAIN 2023

28. (a) Can the absolute refractive index of a medium be less than one?

(b) A coin placed at the bottom of a beaker appears to be raised by 4.0 cm. If the refractive index of water is 4/3, find the depth of the water in the beaker?

MAIN 2023

29. A fish while on the surface of water has its angle of view as α, while it is in water the angle of vision becomes β Compare the two angles of view.

(a) $\alpha = \beta$

(b) $\alpha < \beta$

(c) $\alpha > \beta$

(d) can't say

COMP 2023

30. The image when obtained from a concave lens of focal length f is magnified up to n times the size of the object. What is the distance of the object from the lens ?

(a) $\left(1-\frac{1}{n}\right)f$

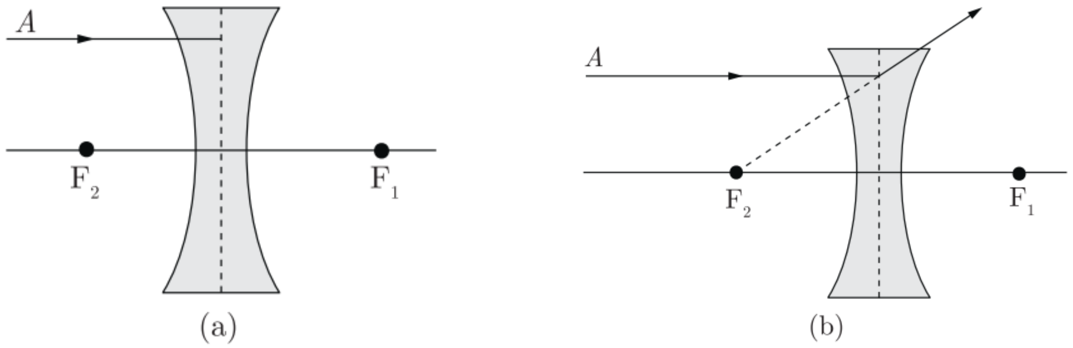
(b) $\left(1+\frac{1}{n}\right)f$

(c) $\left(1-\frac{n}{n}\right)f$

(d) $\left(1+\frac{n}{n}\right)f$

COMP 2023

31. In figure 'a' below of a thin concave lens, F_1 and F_2 are foci. Complete the path of the given ray of light after it emerges out of the lens.



COMP 2023

32. The plano-convex lens has

(a) Both surfaces convex

(b) Both concave

(c) One plane, one convex

(d) One convex, one concave

SQP 2022

33. Where is the object placed in the device shown in the diagram to obtain a highly enlarged size of image?

(a) object placed between F and 2F

(b) object is between O and F

(c) object is placed beyond 2F

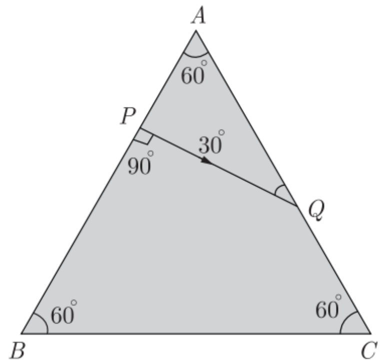
(d) object is placed at F

SQP 2022

34. As the ray of light passes from air to glass, state how the wavelength of light changes. Does it increase, decrease or remain constant?

SQP 2022

35. Complete the path of the ray from Q onwards in the adjoining figure of a glass prism.



SQP 2022

36. A postage stamp appears raised by 7.0 mm when placed under a rectangular glass block of refractive index 1.5. Find the thickness of the glass block.

SQP 2022

37. Can a glass slab disperse light? If not, why?

MAIN 2022

38. Absolute refractive index of some materials A, B, C and D are given as below :

Medium	A	B	C	D
Refractive index	1.54	1.33	2.42	1.65

(a) A

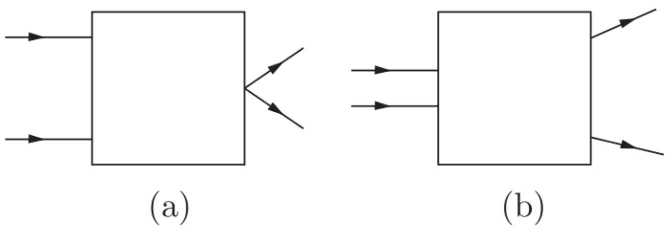
(b) D

(c) C

(d) B

COMP 2022

39. Figures 'a' and 'b' are shown below the incident and transmitted rays through lens kept in a box in each case. Draw the lens and complete the path of rays through it.



COMP 2022

40. The refracted ray is .... to the base of the prism in minimum deviation position.

(a) collinear

(b) parallel

(c) perpendicular

(d) inclined

SQP 2021

41. When a ray of light enters into another optical medium, its wavelength and velocity change. The material in which wavelength and velocity decrease maximum, when the ray is travelling through air is :

(a) glass (b) water

(c) alcohol (d) diamond

SQP 2021

42. Name the waves given out by the two-dish shown in the figure:

(a) Sound waves (b) Light waves

(c) Microwaves (d) Radio waves

SQP 2021

43. If the rays from a point object after refraction through the lens do not actually meet at a point, but they appear to diverge from a point the image is ....

(a) imaginary

(b) beautiful

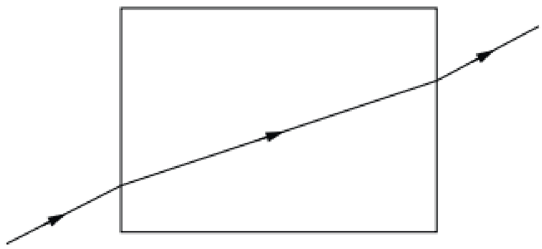
(c) virtual

(d) real

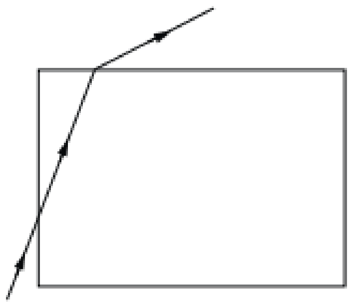
SQP 2021

44. A ray of light is incident on one side of a rectangular glass block. Its path is plotted through the block and out through another side. Which path is not possible ?

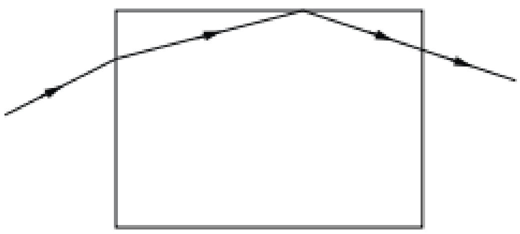
(a)



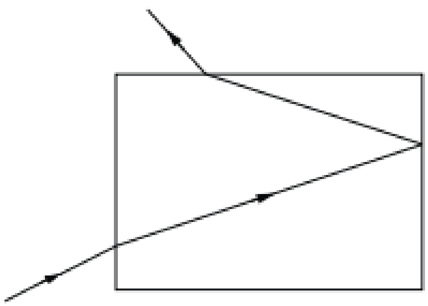
(b)



(c)



(d)



SQP 2021

45. Lateral displacement occurs in case of

(a) rectangular glass block

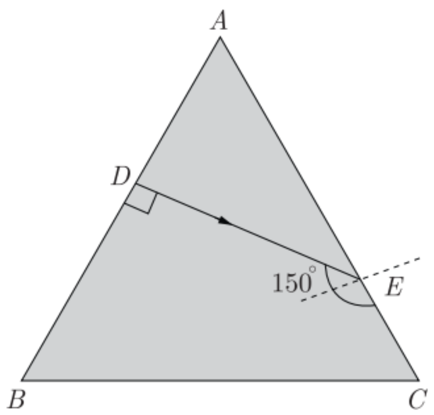
(b) two equiangular prisms held near each other with faces parallel to one another

(c) Cuboid glass block

(d) all of the above

SQP 2021

46. The critical angle for the glass of which the equilateral prism ABC is made is 60°. A ray of light incident on the side AB of the prism is refracted along DE such that the angle it makes with the side AC is 150°. Also, $\angle EDB$ = 90°. Copy the diagram.



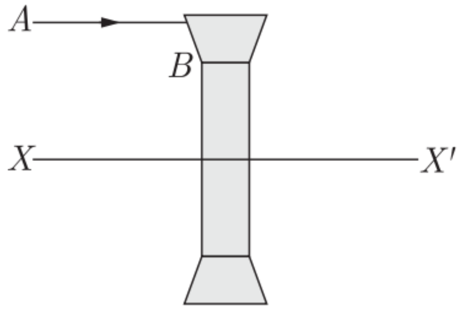
(a) Draw the path of the ray incident on the side AB. (which travels along DE.)

(b) Show the path along which the ray DE travels from the point E onwards and through the side BC.

SQP 2021

47. The diagram shows a lens as a combination of a glass block and two prisms. Name the lens formed by the combination. What is the line XX' called? Complete the path of the incident ray AB after passing through the lens. The final emergent ray will either meet XX' at a point or appear to come from a point on XX'.

What is this point called? Label it in your diagram by the letter F.



SQP 2021

48. During spear fishing a fisherman aims at the :

(a) head of fish

(b) slightly ahead of the head of fish

(c) tail of fish

(d) none of these

MAIN 2021

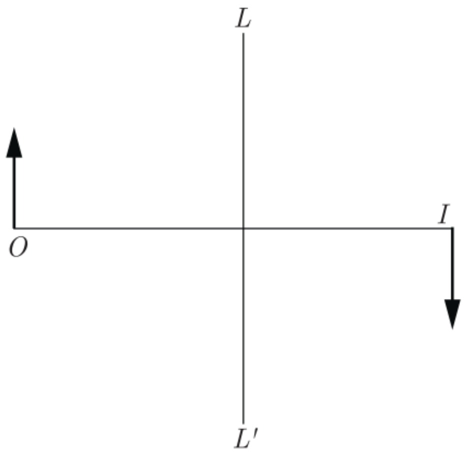
49. Where must an object be placed in front of a converging lens so that the image formed may be

(a) of the same size as the object,

(b) at infinity,

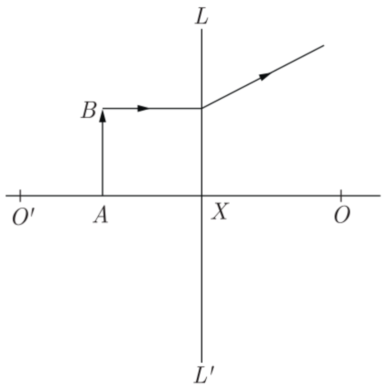
MAIN 2021

50. The diagram shows the object O and the image I formed by a lens. Copy the diagram and on it mark the position of the lens LL' and focus F. Name the lens.



MAIN 2021

51. Study the diagram below and answer the following questions :



(i) Name the lens LL'.

(ii) What are the points O and O' called ?

(iii) Between which points will the image of the object AB be formed ?

(iv) What is the nature of the image ?

MAIN 2021

52. The refractive index of a diamond is 2.4. It means that,

(a) the speed of light in vacuum is equal to $\frac{1}{2.4}$ times the speed of light in diamond

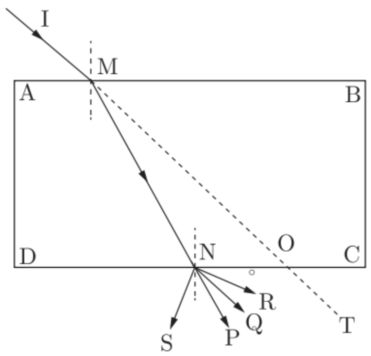
(b) the speed of light in diamond is 2.4 times the speed of light in vacuum

(c) the speed of light in vacuum is 2.4 times the speed of light in diamond

(d) the wavelength of light in diamond is 2.4 times the wavelength of light in vacuum

SEM-I 2021

53. A ray of light IM is incident on a glass slab $ABCD$ as shown in the figure. The emergent ray for this incident ray is



(a) NQ

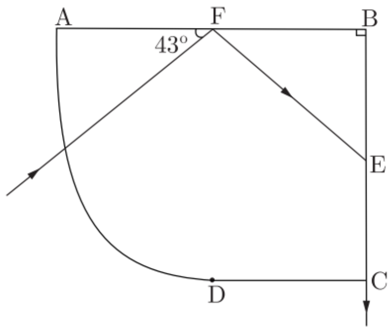
(b) NR

(c) NP

(d) NS

SEM-I 2021

54. A ray of light travelling from air into a glass material as shown below. Answer the questions that follow.



SEM-I 2021

(i) The angle of incidence at the surface AB is

(a) 43°

(b) 47°

(c) 90°

(d) 0°

(ii) Select a correct statement from the following:

(a) The speed of light at the curved surface AD does not change while entering the block.

(b) The ray at the surface AD is not traveling along the radius of the curved part.

(c) The ray at the surface AD is traveling along the radius of the curved part.

(d) Light never refracts when it enters a curved surface.

(iii) The angle of incidence on the surface BC is

(a) 43°

(b) 47°

(c) 90°

(d) 0°

(iv) The critical angle of this material of glass is

(a) 47°

(b) 43°

(c) 42°

(d) 45°

55. The ratio of velocities of light of wavelength 400 nm and 800 nm in a vacuum is

(a) 1 : 1 (b) 1 : 2

(c) 2 : 1 (d) 1 : 3

SEM-I 2021

56. The wavelength range of visible light is

(a) 40 nm to 80 nm

(b) 4000 nm to 8000 nm

(c) 4 nm to 8 nm

(d) 400 nm to 800 nm

SEM-I 2021

57. The deviation produced by an equilateral prism does not depend on the

(a) angle of incidence

(b) size of the prism

(c) material of the prism

(d) colour of light used

SEM-I 2021

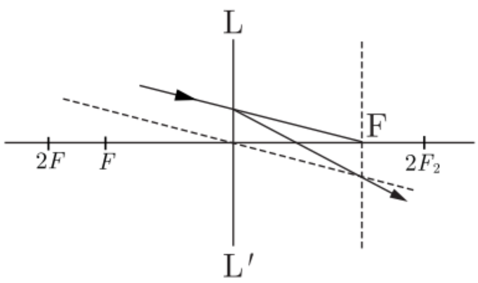
58. The colour of white light which is deviated least by a prism is

(a) green (b) yellow

(c) red (d) violet

SEM-I 2021

59. Observe the diagram which shows the path of an incident ray through an optical plane LL' of a lens. The focal length of the lens in 20 cm.



(i) If an object is placed at a distance of 30 cm in front of this lens, then the image will be

(a) virtual

(b) diminished and inverted

(c) diminished

(d) real and magnified

(ii) This type of lens can be used

(a) to correct hypermetropia

(b) to correct myopia

(c) to diverge light

(d) in the front door peepholes

(iii) An object is placed in front of this lens at a distance of 60 cm, then the image distance from the lens with proper sign convention is

(a) +60 cm

(b) +30 cm

(c) -30 cm

(d) +15 cm

(iv) An object is placed in front of this lens at a distance of 60 cm, then the magnification of the image is

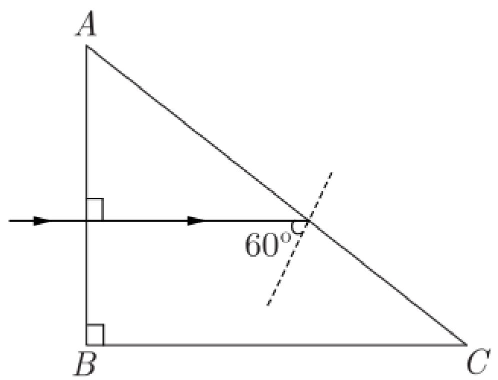
(a) 0.25

(b) 1.25

(c) 0.5

(d) 1

60. The diagram below shows the path of light passing through a right angled prism of critical angle 42°.



The angle C of the prism is

(a) 45° (b) 60°

(c) 90° (d) 30°

SEM-I 2021

61. A thick glass slab with a silvered side forms multiple images on account of :

(a) refraction of light

(b) both reflection and refraction of light

(c) reflection of light

(d) dispersion of light

COMP 2021

62. .... waves are mainly used in radar.

(a) X-rays (b) IR

(c) Radio (d) UV

COMP 2021

Solution

1.

Ans: (d) total internal reflection

Explanation:

Optical fibres guide light through their core by repeated **total internal reflection**, so the light (and image information) travels along the fibre without significant loss.

2.

Ans: (d) slows down less and refracts less

Explanation:

In glass, **red light** has a **higher speed** than blue light, so it is slowed down **less**. Because of this, it is **bent (refracted) less** than blue light inside the prism.

3.

Ans: (b) away from the lens

Explanation:

For a convex lens,

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Here f = 12 cm, u = 20 cm:

$$\frac{1}{v} = \frac{1}{12} - \frac{1}{20} = \frac{5-3}{60} = \frac{2}{60} = \frac{1}{30} \Rightarrow v = 30 \text{ cm}$$

The clear image forms at **30 cm**, but the screen is at **23 cm**, so it must be moved **further away from the lens**.

4.

Ans: 2 images will be formed.

5.

Ans: Concave lens.

6.

Ans: (a)

- **Violet** light shows the **maximum** angle of deviation.
- **Red** light shows the **minimum** angle of deviation.

(b) If a **green** ray passes through a glass prism, the **emergent ray will also be green**, because a prism only disperses white light into different colours, but a single coloured light emerges unchanged.

7.

Ans: (a) Total internal reflection.

(b) 1. Converge

2. Diverge

8.

Ans:

(a) Since the refractive index of water decreases when temperature increases,

$$x < 1.33$$

(b) Differences between normal reflection and total internal reflection (TIR):

Reflection	**Total Internal Reflection (TIR)**
Takes place in any medium, whether denser or rarer.	Takes place only when light travels from a denser to a rarer medium.
Occurs for any angle of incidence.	Occurs only when the angle of incidence is greater than the critical angle.

9.

Ans:

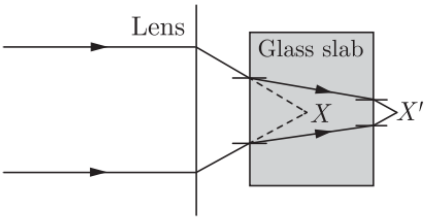
Description	**Label**
(a) an angle of reflection on the mirror	**a**
(b) a partially reflected ray in the glass slab	**P**

(c) a critical angle	**C**
(d) a refracted ray	**Q/T**
(e) an angle of refraction of the ray R	**90°**

10.

Ans: (a) Move away from the slab

Explanation:



When the glass block is inserted, rays inside it bend towards the normal and take a longer optical path. After emerging, they converge at a new point **farther from the slab** than before.

11.

Ans: (a) m < 1

Explanation:

A concave lens always forms a **virtual, erect, and smaller** image of the object.

$$\text{Magnification } m = \frac{\text{height of image}}{\text{height of object}}.$$

Since the image is smaller, $m < 1$.

12.

Ans:

$$\text{Scattering of light} \propto \frac{1}{\text{Wavelength}}$$

The wavelength of ultraviolet radiation is **less** than that of microwave radiation. Therefore, ultraviolet radiations scatter **more** than microwaves.

So, the **Assertion is true**, but the **Reason is false**.

Thus, the correct option is **(b)**.

13.

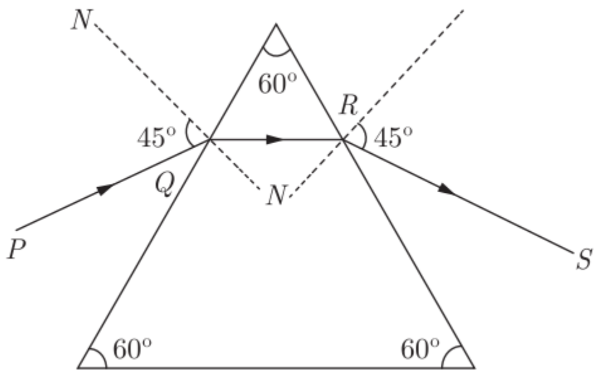
Ans: In between optical centre and focus.

14.

Ans: Because concave lens produces reduced size image.

15.

Ans: (a)



(b) The angle of deviation depends on the following two factors:

1. Angle of incidence

2. Refractive index of the material of the prism

16.

Ans: (a) Refractive index of glass with respect to water = $\frac{9}{5}$

Therefore, the refractive index of water with respect to glass is the reciprocal:

$$\mu_{\text{water/glass}} = \frac{5}{9}$$

(b) The principle used is the **principle of reversibility of light**.

(c) No.

If the temperature of water changes, its refractive index also changes. Hence, the ratio will not remain the same.

17.

Ans: (a) The statement is false.

Light travels slower in a denser medium like diamond. Therefore, it cannot cover a longer distance in the same time compared to vacuum.

(b)

Given:

- Distance in vacuum = $10x$ in time t_1
- Distance in medium = x in time t_2

Speed in vacuum:

$$v_1 = \frac{10x}{t_1}$$

Speed in medium:

$$v_2 = \frac{x}{t_2}$$

Refractive index is:

$$\mu = \frac{v_1}{v_2}$$

$$\mu = \frac{\frac{10x}{t_1}}{\frac{x}{t_2}}$$

$$\mu = \frac{10x}{t_1} \times \frac{t_2}{x}$$

Cancel x:

$$\mu = \frac{10t_2}{t_1}$$

If the question gives values or the ratio simplifies to 3.4, then:

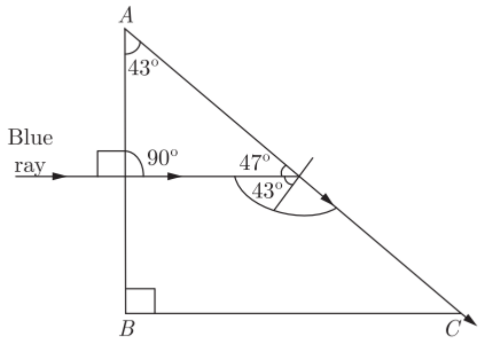
$$\mu = 3.4$$

18.

Ans:

1. Critical angle of the prism material:

The critical angle is 43°.



2. Angle of the prism (∠A):

From the given ray diagram, the angle of the prism is **43°**.

3. Indigo or Violet should replace the blue ray.

Explanation:

Indigo and violet light have refractive indices higher than that of blue light.

Therefore, their critical angles are **less than 43°**.

So, the given incident angle (> 43°) will be greater than their critical angles, causing **total internal reflection**.

19.

Ans:

Using the lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Given:

Object distance, u = −36 cm

Image distance, v = +72 cm

$$\frac{1}{72} - \left(-\frac{1}{36}\right) = \frac{1}{f}$$

$$\frac{1}{72} + \frac{1}{36} = \frac{1}{f}$$

$$\frac{1+2}{72} = \frac{3}{72} = \frac{1}{f}$$

$$f = 24$$

Power of a lens:

$$P = \frac{100}{f(\text{in cm})} = \frac{100}{24} = 4.17\,\text{D}$$

Focal length of the lens = 24 cm

Power of the lens = 4.17 D

20.

Ans: (a) total internal reflection

Explanation:

Optical fibres function on the **principle of total internal reflection**, which allows light to travel through the fibre by repeatedly reflecting within its core without any loss of energy.

21.

Ans:

When a ray of light is incident **normally (i = 0°)** on a plane glass slab:

- The ray enters the glass **without bending**, so the **angle of refraction r = 0°**.
- Since the ray passes straight through without any change in direction, the **angle of deviation is also 0°**.

Therefore:

- **Angle of refraction = 0°**
- **Angle of deviation = 0°**

22.

Ans: Beyond the visible-red end of the electromagnetic spectrum are **infra-red rays**.

These rays have a **longer wavelength** and **lower frequency** than red light.

Use:

Infra-red rays are used in **infra-red cameras** for seeing clearly during fog, at night, or during war to detect enemy objects.

Detection:

Infra-red rays produce heat, so they can be detected using a **thermometer with a blackened bulb**.

When the thermometer is moved from the violet end to the red end of the spectrum, the temperature rises because infra-red rays contain heat energy.

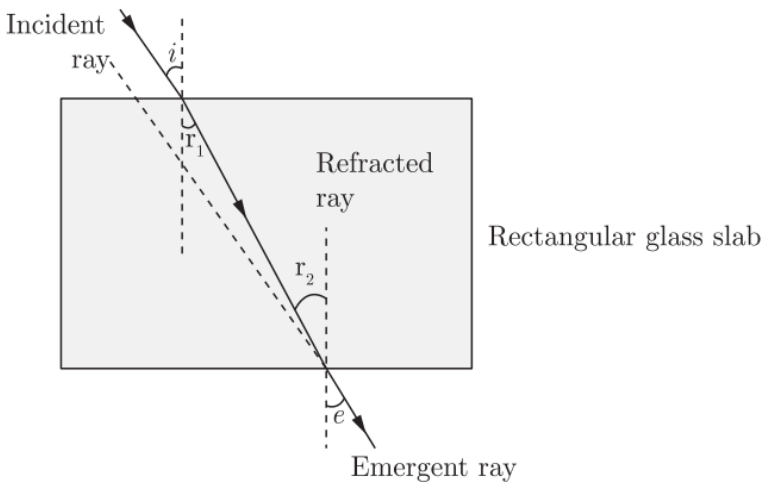
23.

Ans: When a ray of light falls obliquely on a rectangular glass slab of uniform thickness, it bends inside the slab but finally emerges **parallel** to the incident ray.

The **angle of incidence (i)** at the first surface is **equal** to the **angle of emergence (e)** at the second surface.

Pairs of equal angles:

- **Angle of incidence (i) = Angle of emergence (e)**
- **Angle of refraction at first surface = Angle of incidence at second surface**



24.

Ans:

Given:

$P_1 = +5\ D$

$P_2 = -7\ D$

Total power of the combination:

$$P = P_1 + P_2 = (+5D) + (-7D) = -2D$$

Since the **resultant power is negative**, the combination of the two lenses behaves as a **divergent lens**.

25.

Ans: (d) both (b) and (c)

Explanation:

A ray passing through a prism **refracts at each refracting surface**—once on entering and again on emerging. So, it suffers refraction on the first surface (b) and on **both** surfaces (c). (Option (a) is not the safest general statement, but the definite truths are (b) and (c).)

26.

Ans: (b) A convergent and a divergent beam

Explanation:

The parallel beam enters the lens through two different surrounding media:

- **Upper rays:** medium $\mu_2 > \mu_1$ (denser ⟶ rarer at entry) bend **away** from the normal (diverging), then from μ_1 to μ_2 at exit (rarer ⟶ denser) bend **towards** the normal (converging). Net effect: **convergent**.
- **Lower rays:** medium $\mu_3 < \mu_1$ (rarer ⟶ denser at entry) bend **towards** the normal (converging), then from μ_1 to μ_3 at exit (denser ⟶ rarer) bend **away** (diverging). Net effect: **divergent**.

So, the lens produces **one convergent and one divergent beam**.

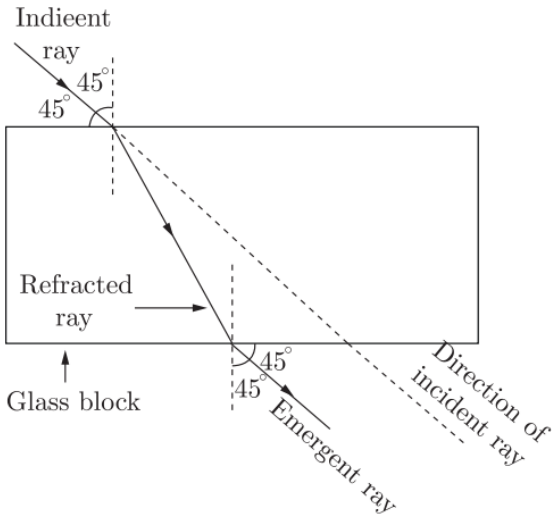
27.

Ans: The emergent ray will come out of the glass slab **parallel to the incident ray** (but laterally shifted). This happens due to **Snell's law of refraction**.

The diagram should show:

- The incident ray entering the slab at 45°.
- The ray bending towards the normal inside the slab.

- The emergent ray bending away from the normal and coming out **parallel** to the incident ray.



28. (a)

Ans: No. The absolute refractive index of a medium is given by

$$\mu = \frac{c}{v},$$

where *c* is the speed of light in vacuum and *v* is the speed of light in the medium.

Since light travels fastest in vacuum (*c* > *v* for all media),

$$\mu > 1.$$

Hence, the refractive index of a medium cannot be less than one.

28. (b)

Ans: Let the real depth of the coin be x.

Apparent depth = real depth – raised amount

$$\text{Apparent depth} = x - 4$$

Refractive index of water:

$$\mu = \frac{\text{real depth}}{\text{apparent depth}}$$

$$\frac{4}{3} = \frac{x}{x-4}$$

Cross-multiplying:

$$4(x - 4) = 3x$$

$$4x - 16 = 3x$$

$$x = 16 \text{ cm}$$

Real depth of water = 16 cm

29.

Ans: (c) $\alpha > \beta$

Explanation:

Refraction at the water–air boundary gives

$$\frac{\sin\alpha}{\sin\beta} = \frac{n_{\text{water}}}{n_{\text{air}}} > 1.$$

Hence $\sin \alpha > \sin \beta$ and therefore $\alpha > \beta$.

So, the angle of view at the surface is **larger** than when the fish is in water.

30.

Ans: (c) $\left(1 - \frac{n}{n}\right) f$

Explanation:

Lens formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

Magnification for a lens: $m = \frac{v}{u} = n \Rightarrow v = nu$.

Substitute $v = nu$:

$$\frac{1}{nu} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1-n}{nu} = \frac{1}{f} \Rightarrow u = \left(\frac{1-n}{n}\right) f.$$

31.

Ans: Figure (b) shows the complete path of the emergent ray after passing through the concave lens.

The ray, after refraction through the concave lens, diverges as if it is coming from the focus F_1.

32.

Ans: (c) One plane, one convex

Explanation:

A **plano-convex lens** has **one flat (plane)** surface and **one outward curved (convex)** surface.

This type of lens helps to **converge light rays** to a single point.

Hence, option **(c)** is correct.

33.

Ans: (b) object is between O and F

Explanation:

The microscope uses a **convex lens** to get a highly enlarged image. When an object is placed **between the optical centre (O) and the focal point (F)** of a convex lens, it forms a **virtual, erect, and highly magnified** image.

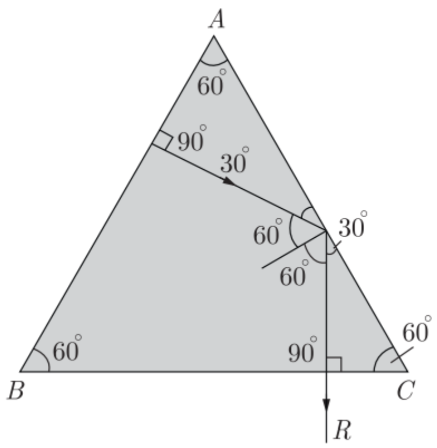
Hence, option **(b)** is correct.

34.

Ans: When a ray of light passes from air into glass, its wavelength λ **decreases**.

35.

Ans:



36.

Ans:

Let the thickness of the glass block be ***t* mm**.

A postage stamp appears raised by **7 mm**, so:

Apparent thickness = (t − 7) mm

Refractive index of glass:

$$\mu = \frac{\text{Real thickness}}{\text{Apparent thickness}} = \frac{t}{t-7}$$

Given,

$$\mu = 1.5$$

So,

$$1.5 = \frac{t}{t-7}$$

Cross-multiplying:

$$1.5(t - 7) = t$$

$$1.5t - 10.5 = t$$

$$0.5t = 10.5$$

$$t = \frac{10.5}{0.5} = 21 \text{ mm}$$

Therefore, the **thickness of the glass block = 21 mm = 2.1 cm**.

37.

Ans: No, a glass slab cannot show dispersion of light.

Although white light splits into different colours inside the slab, these colours **recombine** when they emerge from the other parallel face.

Therefore, **the emergent light becomes white again**, so no dispersion is seen.

38.

Ans: (d) B

Explanation:

Light travels faster in a medium that has a **lower refractive index** because such a medium is **optically rarer**.

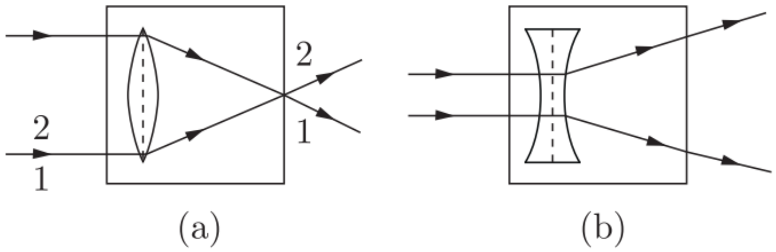
From the table:

- A = 1.54
- **B = 1.33**
- C = 2.42
- D = 1.65

Since **B has the smallest refractive index (1.33)**, light travels **fastest** in medium **B**.

39.

Ans: The ray diagram is as follows :



40.

Ans: (b) parallel

Explanation:

When the **refracted ray** inside a prism is **parallel to the base**, the **angle of deviation** becomes **minimum**.

At this position, the **angle of incidence** is equal to the **angle of emergence**.

Hence, option **(b)** is correct.

41.

Ans: (d) diamond

Explanation:

When light passes from air into another medium, its **speed and wavelength decrease**.

The **denser** the medium, the **greater** is the decrease.

Since **diamond** is the **densest optical medium** among the given options, the **wavelength and velocity decrease the most** in diamond.

42.

Ans: (d) Radio waves

Explanation:

Communication towers send and receive **long-range signals** as **radio waves**, which can travel large distances through the atmosphere.

Hence, option **(d)** is correct.

43.

Ans: (c) virtual

Explanation:

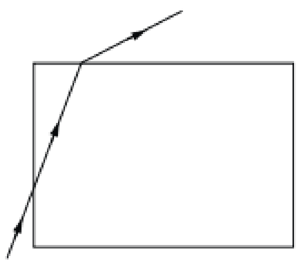
If the refracted rays **do not actually meet** but only **appear to come from** a point when traced backward, the image is **virtual**.

A virtual image **cannot be obtained on a screen** (e.g., image in a plane mirror or by a concave lens).

44.

Ans: (b)

Explanation:



At air → glass (denser) the ray **bends towards the normal**; at glass → air (rarer) it **bends away from the normal**.

Path (b) shows the emerging ray bending the wrong way, so it's **not possible**.

45.

Ans: (d) all of the above

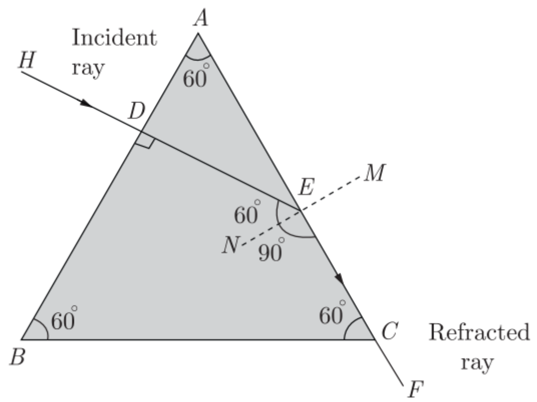
Explanation:

Lateral displacement happens when the **opposite faces of the medium are parallel**.

In all the given cases — rectangular glass block, two prisms with parallel faces, and cuboid glass block — the **lateral surfaces are parallel**, causing the light to emerge **displaced but parallel** to the incident ray.

46.

Ans: (a) Since the ray DE inside the prism is traveling **normally to side AB**, the angle of incidence on AB is **0°**. Hence it passes straight through without bending.



(b) Given: $\angle EDB$ = 90° and $\angle ACD$ = 150°.

Using the geometry of the prism:

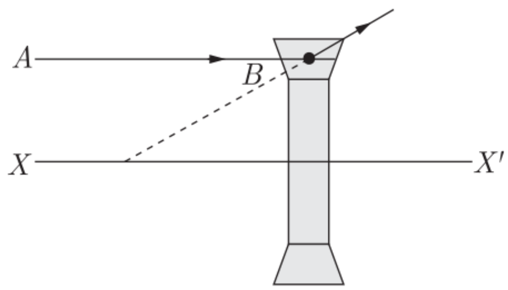
$$\angle DEN = \angle DEC - \angle NEC = 150° - 90° = 60°$$

The angle of refraction inside the prism becomes **60°**, which is equal to the **critical angle** of the glass.

Therefore, the emergent ray will travel **along EC**, undergoing **total internal reflection** at point E and moving toward CF.

47.

Ans: The lens formed by the combination is a concave lens. The line XX' is called the principal axis. The completed ray diagram is as shown in figure below. The emergent rays appears to be coming from a point F on the line XX'. This point is called the second focus of the lens.



48.

Ans: (c) tail of fish

Explanation:

Light from the fish goes from **water to air** and **refracts**, so the fish appears **higher and slightly forward** compared to its real position. If the fisherman aims at the apparent head, the spear will pass **above/forward** of the real head. Aiming at the **tail of the apparent image** compensates for this shift and hits the **actual head** of the fish.

49.

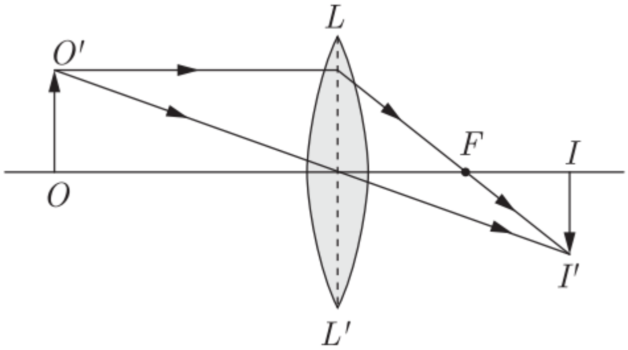
Ans: (a) The object must be placed **at 2F** to get an image of the **same size** as the object.

(b) The object must be placed **at F** so that the image is formed **at infinity**.

50.

Ans: The lens shown is a **convex lens**.

A convex lens forms a **real, inverted image** on the opposite side of the lens, as shown in the diagram. The lens LL' should be placed vertically between the object O and the image I, and its focus F is marked on the principal axis.



51. (i)

Ans: The lens LL' is a concave lens.

51. (ii)

Ans: The points O and O' are the first and second focal points, respectively.

51. (iii)

Ans: The image of object AB will be formed between A and X.

51. (iv)

Ans: The image formed is **diminished, virtual, and erect**.

52.

Ans: (c) the speed of light in vacuum is 2.4 times the speed of light in diamond

Explanation:

Refractive index n is defined as

$$n = \frac{\text{speed of light in vacuum } (c)}{\text{speed of light in the medium } (v)}.$$

For diamond, $n = 2.4 \Rightarrow \frac{c}{v} = 2.4 \Rightarrow c = 2.4v$.

So light travels **2.4 times faster in vacuum** than in diamond.

53.

Ans: (a) NQ

Explanation:

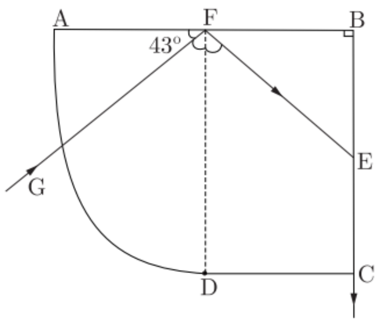
When a ray of light passes through a glass slab, it bends towards the normal while entering and away from the normal while emerging. The emergent ray is **parallel to the incident ray** but **displaced sideways**.

Hence, in the figure, $\boldsymbol{NQ}$ is the emergent ray.

54. (i)

Ans: (b) 47°

Explanation:



At the reflecting surface, **angle of incidence = angle of reflection**.

From the figure, the given 43° is complementary to the angle made with the **normal**, so

$$i = 90° - 43° = 47°.$$

Hence, the angle of incidence at AB is **47°.**

54. (ii)

Ans: (c) The ray at the surface AD is traveling along the radius of the curved part.

Explanation:

At the curved surface AD, the ray passes along the **radius** of curvature.

A ray traveling along the radius strikes the surface **normally**, so it **does not refract** and continues in the same direction.

54. (iii)

Ans: (a) 43°

Explanation:

By the law of reflection at surface BC, **angle of incidence = angle of reflection**.

From the diagram, the reflected angle is **43°**, so the **angle of incidence on BC** is also **43°.**

54. (iv)

Ans: (a) 47°

Explanation:

At the glass–air boundary, the emergent ray makes **43°** with the **normal**, so it makes **47°** with the **surface**.

The **critical angle** is the angle of incidence in glass that produces a **90°** refracted angle (i.e., along the surface), equal to the angle the emergent ray makes **with the surface** here.

Hence, θ_c = 47°.

55.

Ans: (a) 1 : 1

Explanation:

All electromagnetic waves travel at the **same speed in vacuum** (3×10^8 m/s), independent of wavelength.

So, light of 400 nm and 800 nm both have the **same velocity**, giving a ratio **1 : 1**.

56.

Ans: (d) 400 nm to 800 nm

Explanation:

The approximate wavelength range of **visible light** is **400 nm (violet)** to **800 nm (red)**.

Hence, option **(d)** is correct.

57.

Ans: (b) size of the prism

Explanation:

Angle of deviation depends on **angle of incidence**, **refractive index** (hence material and colour/wavelength), and the **apex angle** of the prism—not on its physical **size**. For an equilateral prism (apex 60°), changing size doesn't change deviation.

58.

Ans: (c) red

Explanation:

In a prism, **deviation increases as wavelength decreases**.

Red light has the **longest wavelength** among visible colours, so it is **deviated the least**.

Hence, option **(c)** is correct.

59. (i)

Ans: (d) real and magnified

Explanation:

For the given convex lens,

Focal length, f = +20 cm

Object distance, u = −30 cm

Using the **lens formula**:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{30}$$

$$\frac{1}{v} = \frac{1}{60}$$

So,

$$v = +60 \text{ cm}$$

Since v is positive, the image is **real** and formed on the opposite side of the lens.

Also, the image is formed farther away from the lens than the object ($v > u$), so the image is **magnified**.

Hence, the image is **real and magnified**.

59. (ii)

Ans: (a) to correct hypermetropia

Explanation:

This lens is a **convex lens**. Convex lenses converge light and are used to correct **hypermetropia (long-sightedness)** because they help the eye focus light properly on the retina.

So, the correct option is **(a)**.

59. (iii)

Ans: (b) +30 cm

Explanation:

Object distance is **u = –60 cm** (negative because the object is in front of the lens).

Focal length of the convex lens is f = **+20 cm**.

Using the lens formula:

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{60}$$

$$\frac{1}{v} = \frac{3-1}{60} = \frac{2}{60}$$

$$v = +30 \text{ cm}$$

Since v **is positive**, the image is formed on the other side of the lens.

So, the image distance is **+30 cm**.

Thus, **(b)** is correct.

59. (iv)

Ans: (c) 0.5

Explanation:

Object distance:

$$u = -60 \text{ cm}$$

Image distance:

$$v = +30 \text{ cm}$$

Magnification is given by:

$$m = -\frac{v}{u}$$

$$m = -\frac{+30}{-60} = 0.5$$

So, the image is **half the size of the object**.

Thus, **(c)** is the correct option.

60.

Ans: (d) 30°

Explanation:

From the geometry of the right-angled prism, the angle at the vertex C equals $\angle AOP$.

Given the shown internal angle is 60°, so

$$\angle C = \angle AOP = 90^\circ - 60^\circ = 30^\circ.$$

Hence, the prism angle C is **30°**.

61.

Ans: (b) both reflection and refraction of light

Explanation:

A thick glass slab with one side silvered behaves like a thick mirror. When light falls on it, part of the light is **reflected** from the surfaces and part is **refracted** into the glass, then reflected again from the silvered side and comes out.

Because these reflections and refractions happen at **both surfaces**, several rays come out, each forming a separate image.

So, multiple images are formed due to **both reflection and refraction of light**.

62.

Ans: (c) Radio

Explanation:

Radar systems use **radio waves** because these waves can travel long distances, reflect well from objects, and are not easily absorbed by the atmosphere. Hence, radio waves are mainly used in radar.